V **SEMESTER**	L	T	P	C
(JOB ORIENTED ELECTIVE-I)	2	-	2	3
20IT5J01 - LINUX ADMINISTRATION				

COURSE OUTCOMES:

At the end of the course, the students will be able to:

CO1: Use various Linux commands that are used to manipulate system operations at admin level.
CO2: Write Shell Programming using Linux commands.
CO3: Design and write application to manipulate internal kernel level Linux File System.
CO4: Explain the user, group and storage management.
CO5: Configure SSH client and server.

UNIT -I:

Introduction To Linux And Linux Utilities: A brief history of LINUX, architecture of LINUX, features of LINUX, introduction to vi editor. Linux commands- PATH, man, echo, printf, script, passwd, uname, who, date, stty, pwd, cd, mkdir, rmdir, ls, cp, mv, rm, cat, more, wc, lp, od, tar, gzip, unlink, du, find, unmask, ulimit, ps, finger, tail, head, sort, nl, uniq, cut, paste, join, tee, pg, comm, cmp, diff, tr, awk, cpio, apt.

UNIT -II:

Introduction to Shells: Linux Session, Standard Streams, Redirection, Pipes, Tee Command, Command Execution, Command-Line Editing, Quotes, Command Substitution, Job Control, Aliases, Variables, Predefined Variables, Options, Shell/Environment Customization.
Filters: Filters and Pipes, Concatenating files, Display Beginning and End of files, Cut and Paste, Sorting, Translating Characters, Files with Duplicate Lines, Count Characters, Words or Lines, Comparing Files.

UNIT -III:

Grep: Operation, grep Family(grep, egrep, fgrep), Searching for File Content.
Sed: Scripts, Operation, Addresses, commands, Applications, grep and sed.
Unix File Structure: Introduction to UNIX file system, inode (Index Node), file descriptors, system calls and device drivers.

UNIT -IV:

User and Group Management: User accounts, local groups and group memberships, Configure networking and hostname resolution statically or dynamically, start, stop, and check the status of network services and network related commands.
Storage Management: List, create, delete, and modify physical storage partitions and tools

UNIT -V:

Configuring SSH: Enabling the SSH Server, Using the SSH Client, Configuring Key- Based SSH Authentication, Using Graphical Applications with SSH.
Practical Learning: Installation of Any open source Linux Distribution, AWS Instance Creation and Learn How to Access through SSH.

Text Books:

1. W. Richard. Stevens, Advanced Programming in the UNIX Environment, 3rd edition,
Pearson Education, New Delhi, India.
2. Behrouz A. *Forouzan, Richard F. Gilberg*, Unix and shell Programming Thomson

Reference Books:

1. *Robert Love, O'Reilly*, Linux System Programming, SPD.
2. *W.R.Stevens*, Advanced Programming in the UNIX environment, 2nd Edition, PearsonEducation.
3. *W.R. Stevens*, UNIX Network Programming, PHI.
4. *Graham Glass, King Ables*, UNIX for Programmers and Users, 3rd Edition, Pearson Education.